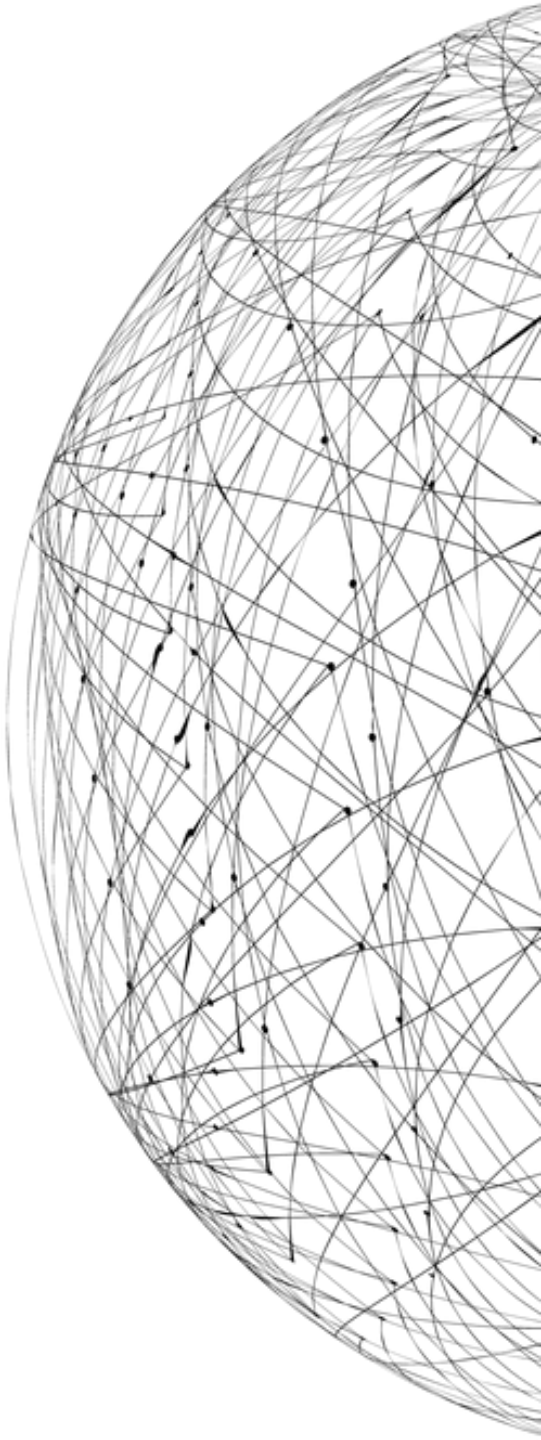
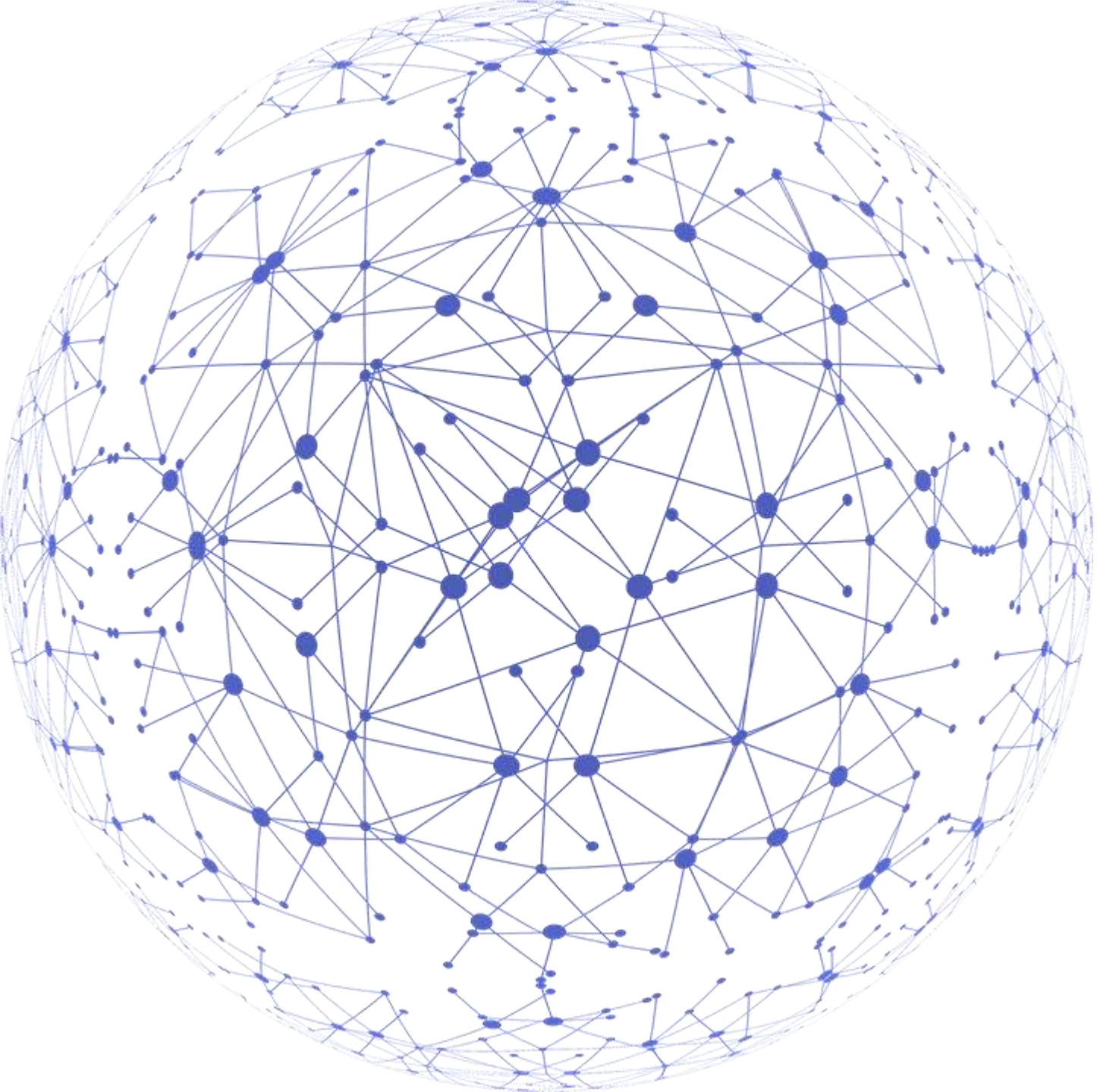
Internet Programming

Chapter 2

Web Page Development Using HTML





Content

- What is HTML
- Creating HTML documents
- How to specify colors?
- Text Formatting in HTML

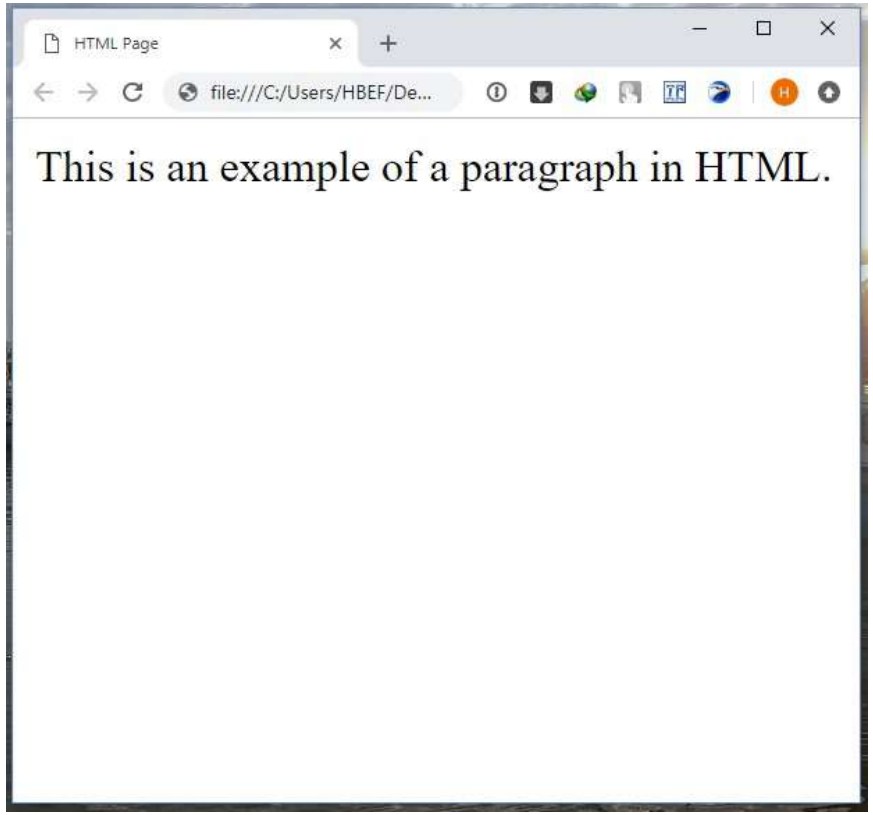
What is HTML (cont...)

- HTML stands for (Hypertext Markup Language)
- A **markup language** is a computer language that uses tags to define elements within a document. It is human-readable, meaning markup files contain standard words, rather than typical programming syntax. While several markup languages exist, the two most popular are HTML and XML.

What is HTML (cont…)

- HTML stands for "Hypertext Markup Language“. It is the language used to create webpages. "Hypertext" refers to the hyperlinks that an HTML page may contain. "Markup language" refers to the way tags are used to define the page layout and elements within the page.
- An open standard
  - Developed by the World Wide Web Consortium (W3C)

```
<html>
<head>
<title>HTML Page</title>
</head>
<body>
<p>This is an example of a paragraph in HTML.</p>
</body>
</html>
```



What is HTML (cont…)

- Static—it is only display.
  - Use other technologies to add dynamic function:
    - Client-side (browser) scripting languages (JavaScript)
    - Server-side programs (PHP, Java servlets)
- HTML is a lightweight file format.
  - Size of file containing just “Hello World!”:

File Type	Size
PDF	4,915 bytes
MS Word	19,456 bytes
Postscript	11,274 bytes
HTML	12 – 50 bytes

What is HTML (cont…)

- HTML5
  - is the latest evolution of the standard that defines HTML. The term represents two different concepts. It is a new version of the language HTML, with new elements, attributes, and behaviors, and a larger set of technologies that allows the building of more diverse and powerful websites and applications. This set is sometimes called HTML5 & friends and often shortened to just HTML5.
  - HTML5 is a markup language used for structuring and presenting content on the World Wide Web. It is the fifth and current major version of the HTML standard.

Creating HTML documents

- What are HTML tags?
  - When a web browser displays a page, it reads from a text file, and looks for special codes or "tags" that are marked by the < and > signs. The general format for a HTML tag is: *<tag name>string of text</tag name>*
  - *Example:*

    <h3>What are HTML tags?</h3>

  This tag tells a web browser to display the text ‘What are HTML tags?’ in the style of header level 3. HTML tags may tell a web browser to bold the text, italicize it, make it into a header, or make it be a hypertext link to another web page.

Creating HTML documents (cont...)

- What are HTML tags? (cont...)
  - It is important to note that the ending tag, *</tag name>* contains the "/" slash character. This "/" slash tells a web browser to stop tagging the text. Many HTML tags are paired this way. If you forget the slash, a web browser will continue the tag for the rest of the text in your document, producing undesirable results.
  - Unlike computer programming, if you make a typographical error in HTML you will not get a "bomb" or "crash" the system; your web page will simply look, well... wrong.

Creating HTML documents (cont…)

- An HTML document contains two distinct parts, the head and the body. The head contains information about the document that is not displayed on the screen. The body then contains everything else that is displayed.
- The basic structure then of any HTML page is:

```
<html>
  <head>
    ** [header info used by some web servers, not displayed] **
  </head>

  <body>
    ** [all the HTML for display] **
        :     :
        :     :
  </body>
</html>
```

Creating HTML documents (cont…)

- <HTML>……</HTML>
  - These tags begin and end an HTML document
  - Used to bracket an entire HTML document.
  - Optional for most browsers.
  - Attributes:
    - lang = language code
  - Language code indicates the primary language of the document.
    - en = English

Creating HTML documents (cont...)

- <HEAD >……</HEAD>
  - These tags are in the beginning of the document.
  - Important information is stored in-between these tags including: title, meta-data, styles, and programming scripts.
  - Used to provide information about a web page.
  - Nests within itself other tags like:

    <base>, <meta> and <title>

Creating HTML documents (cont...)

- <base>
  - Attribute: href = url
  - Specifies the base pathname for all relative URLs in the document.
  - No end tag.
- <meta>
  - Used to provide information about a document.
  - Keywords or descriptions to aid search engines.
- < title >……</ title >
  - These tags are in-between the HEAD tags and contain the text that appears in the title of the webpage.
  - Usually appears on the title bar of the browser window.

Creating HTML documents (cont…)

- <BODY>…..</BODY>
  - alink = color
    - Sets the color of active links.
  - link = color
    - Sets the default color for all the links in a document.
  - vlink = color
    - Sets the color for the visited links.
  - bgcolor = color
    - Sets the color for the entire page.

Creating HTML documents (cont...)

- After you wrote your code on any text-file editor tool, save it with an extension of ***htm*** or ***html***.
- And open it with any browser. Your browser will display the contents based on what you (the programmer) tell it to.

How to specify colors?

1. By specifying the color name.
   - Some of the colors that are supported by browsers are:

aqua	black	blue	fuchsia
gray	green	lime	maroon
Navy	olive	purple	red
Silver	teal	yellow	white

   - Many other colors are possible.
   - Example:

```
<body text="white">
<body bgcolor="yellow">
```

How to specify colors? (cont...)

2. By specifying the red, green, blue or RGB components.
   - Each color encoded in 8 bits.
   - 00 means that the color is turned off; FF means fully turned on.
   - Example:

     <body text="#FFFFFF" bgcolor="#0000FF">

Text Formatting in HTML

Paragraphs and Headings

- <Hn> ……. </Hn>
  - Used to generate headings, $1 \leq n \leq 6$.
  - Six different levels of headings.
    - <H1> is the largest, <H6> is the smallest.
- <P>
  - Paragraph marker, used to separate text into paragraphs.
  - End tag </P> is optional.
  - A series of paragraph tags <p><p>…<p> with no intervening text is treated as a single <p>.

Text Formatting in HTML (cont...)

-
  - Used to indicate that the text following <BR> should begin on the next line.
  - The amount of line spacing is less than that of a <P> break.
  - Example:

    This is the first line.

    This is the second line.

    This is the third.

Text Formatting in HTML (cont...)

- <HR>
  - Produces a horizontal line, which can be used to delimit sections.
  - Length of the line can be specified.
  - Examples:

```
<hr>
<hr size= "20"> [ noshade option possible ]
<hr width= "75%">
<hr align="right" width=120>
```

    - Across full width of browser, 20 pixels thick, 75% of available page width, and 120 pixels right-justified.

Text Formatting in HTML (cont…)

- <address> ……. </address>
  - Supplies the contact information of the author.
  - Generally formatted in italics, with a line break above and below.
  - Example

```
<address>
   Mr  X Y<br>
   Dept. of Information Systems <br>
   Bahir Dar University <br>
   Email:  xyz@gmail.com
</address>
```

Text Formatting in HTML (cont...)

- …….
  - Displays the enclosed text in bold.
- …….
  - Displays the enclosed text in italics.
- <u>…..</u>
  - Under line the enclosed text
- <cite> ……. </cite>
  - Tells the browser that this is a citation. Usually displayed in italics.

Text Formatting in HTML (cont...)

- <pre> ……. </pre>
  - Allows browser to display carefully laid out text.
  - Used to display program listings.
  - Example:

```
<pre>
   main()
    {
      int i, j;
      abc ();
     }
</pre>
```

Text Formatting in HTML (cont...)

- _{…….}
  - Displays the enclosed text as subscript.
- ^{…….}
  - Displays the enclosed text as superscript.
- <font> ……. </font>
  - Specifies the style of the enclosed text.
  - Attributes:
    - color = color name
    - face = typeface
    - size = value [1 to 7; 3 is the default]

Text Formatting in HTML (cont...)

- …….
  - Centers the enclosed elements horizontally on the page.
- <P align=option> ……. </P>
  - Used to align a paragraph.
  - Option can be left, right or center.

Example 1

```
<html>
   <head>
         <title> Title of the Document
         </title>
   </head>
   <body  text="white" bgcolor=  "blue">
         This is the content of the
         document. This is an <i> italic </i>
         font,
         and this is <b> bold </b>.
   </body>
</html>
```

Example 2

```
<html>
   <head>
         <title>My First Webpage</title>
   </head>
   <body>
   <h1 align="center">My First Webpage</h1>
   <p>Welcome to my first web page. I am
   writing this page using a text editor </p>
   <p>By learning html, I'll be able to create
   web pages like....<br> which I am of
   course.</p>
   </body>
</html>
```

Example 4

```
<html>
    <head><title>Paragaph Aligning</title></head>
    <body text=white bgcolor=green>
    <h3>
    <P ALIGN=CENTER> This paragraph will be aligned at
    the center. Even as the browser window size changes,
    the alignment remains the same. </P>
    <P ALIGN=LEFT>This demonstrates left alignment.
    </P>
    <P ALIGN=RIGHT>How about aligning by the right
    margin? </P>
    </h3>
    </body>
</html>
```

Example 3

```
<html>
    <head><title>Demonstration of Headings
    </title></head>
    <body text="#FFFFFF" bgcolor="#0000FF">
            <h1>This is a first level heading. </h1>
            <h2>The second level</h2>
            <h3>The third level</h3>
            <h4>Fourth level. </h4>
            <h5>Fifth level.</h5>
            <h6>And, finally, the sixth .</h6>
            A small amount of plain non-heading text.
    </body>
</html>
```

Example 5

```
<html>
   <head><title>Layout Features 1</title></head>
   <body text=yellow bgcolor=blue>
         <h2> <pre> begin
           if (a > b)
                  then max := a; else max
                  := b;
           end;
         </pre> </h2>
         <hr size=8 width=50%>
         <hr>
         <hr size=20 width="75%" noshade>
   </body>
</html>
```

Example 6

```
<html>
   <head><title> Superscript and Subscript
   </title></head>
   <body text=white bgcolor=blue>
         <h1> y = x <sup> 3 </sup> + 2 x
         <sup> 2 </sup> + 4 </h1>
         <br>
         <h2> W <sub> total </sub> = x
         <sup> 2 </sup> - 5 </h2>
   </body>
</html>
```

Lists

Types of Lists

- There are a number of tags for building lists.
  - Serves the purpose of improving the readability of the text.
- Depending on the way the list items are displayed, lists may be divided into three types:
  - Unnumbered list
  - Numbered list
  - Definition list

Lists (cont...)

Unnumbered List

- Used to display a set of related items that appear in no particular order.
  - Each item is indented with a preceding bullet symbol.
- Specified using the tag:

  <UL> …… </UL>
- The individual items in the list are specified using the <LI> tag.
- Attributes:

  type = disc | circle | square

Lists (cont...)

Unnumbered List (cont...)

- The <li> items can contain multiple paragraphs, specified using <p>.
- Example:

```
<ul>
    <li> First item of the list
    <li> Second item of the list
    <li> Third item of the list
</ul>
```

Lists (cont...)

Unnumbered List (Example 1)

```
<html>
    <head><title> Bulleted list 1 </title></head>
    <body text=white bgcolor=blue>
        <h2> The flowers I like: </h2>
        <h3>
            <UL>
                <LI> Rose
                <LI> Lotus
                <LI> Daffodil
                <LI> Marigold
            </UL>
        </h3>
    </body>
</html>
```

Lists (cont...)

Unnumbered List (Example 2)

```
<html>
    <head><title> Bulleted list 2 </title></head>
    <body text=white bgcolor=blue>
        <h2> The flowers I like: </h2>
        <h3>
        <UL type=square>
            <LI> Rose
            <LI> Lotus
            <LI type=circle> Daffodil
            <LI> Marigold
        </UL>
        </h3>
    </body>
</html>
```

Lists (cont...)

Numbered List

- Numbered or ordered lists are used when the sequence of the items is important.
- Specified using the tag:

  <OL> …… </OL>
- The individual items in the list are specified using the <LI> tag.
- Example:

```
<OL>
    <LI> This is number one
    <LI> This is number two
    <LI> This is number three
</OL>
```

Lists (cont...)

Numbered List (cont...)

- The list items are numbered sequentially as 1,2,3,…
- Attributes:
  - type = 1 | A | a | I | i
  - Specifies the style of numbering.
    - 1,2,3,… or A,B,C,… or a,b,c,… or I,II,III,… or i,ii,iii,…

Lists (cont...)

Numbered List (Example 1)

```
<html>
    <head><title> Numbered list 1 </title></head>
    <body text=white bgcolor=blue>
        <h2> To cook Maggi you must: </h2>
        <h3>
        <OL>
            <LI> Put some water to boil
            <LI> Put in noodles and masala
            <LI> Wait for 2 minutes
            <LI> Serve in a plate
        </OL>
        </h3>
    </body>
</html>
```

Lists (cont…)

Numbered List (Example 2)

```
<html>
    <head><title> Numbered list 2 </title></head>
    <body text=white bgcolor=blue>
        <h2> To cook Maggi you must: </h2>
        <h3>
        <OL type =A>
            <LI> Put some water to boil
            <LI> Put in noodles and masala
            <LI> Wait for 2 minutes
            <LI> Serve in a plate
        </OL>
        </h3>
    </body>
</html>
```

Lists (cont...)

Numbered List (Example 3)

```
<html>
   <head><title> Numbered list 3 </title></head>
   <body text=white bgcolor=blue>
      <h2>To cook Maggi you must: </h2>
      <h3>
      <OL type =I>
         <LI> Put some water to boil
         <LI> Put in noodles and masala
         <LI type=1> Wait for 2 minutes
         <LI> Serve in a plate
      </OL>
      </h3>
   </body>
</html>
```

Lists (cont...)

Definition List

- Specified using the tag:

  <DL> …… </DL>

- Each definition comprises of a definition term (<DT>) and a definition description (<DD>).
- Web browsers format each definition on a new line and indent it.

Lists (cont...)

Definition List (Example)

```
<html>
    <head><title> Definition list 1 </title></head>
    <body text=white  bgcolor=blue>
        <h2> Some important protocols: </h2>
        <h3>
        <DL>
          <DT> TCP
             <DD> Transmission Control Protocol
          <DT> UDP
             <DD> User Datagram Protocol
         <DT> IP
             <DD> Internet Protocol
        </DL>
        </h3>
    </body>
</html>
```

Lists (cont...)

Nesting of Lists

- Any list can be nested within another list.
  - When unnumbered lists are nested, the browser automatically uses a different bullet symbol for each level.
  - When numbered lists are nested, by default, every level is numbered using the Arabic numerals (1, 2, 3, …).

Lists (cont...)

Nesting of Lists (Example 1)

```
<html><head><title> List Nesting 1 </title></head>
    <body text=white bgcolor=blue>
        <H3> My favorite languages are:
        <UL>
        <LI> Java <BR>
            <UL>
                <LI> object-oriented
                <LI> platform independent
                <LI> can be embedded within HTML
            </UL> <BR>
        <LI> Perl <BR>
            <UL>
                <LI> a scripting language
                <LI> powerful string handling capabilities
                <LI> widely used for writing CGI scripts
        </UL>
    </UL> </H3>
</body></html>
```

Lists (cont...)

Nesting of Lists (Example 2)

```
<html><head><title> List Nesting 2 </title></head>
    <body text=white  bgcolor=blue>
    <H3> My favorite languages are:
    <OL>
    <LI> Java <BR>
      <UL>
          <LI> object-oriented
           <LI> platform independent
           <LI> can be embedded within HTML
       </UL> <BR>
    <LI> Perl <BR>
       <OL>
           <LI> a scripting language
           <LI> powerful string handling capabilities
           <LI> widely used for writing CGI scripts
        </OL>
    </OL> </H3>
</body></html>
```

Linking between pages

Hyperlinks

- Chief power of HTML comes from its ability to link text and/or image to another document or section of a document.
- These links are called hyperlinks.
  - Browser by default highlights the hyperlinks with color and/or underline.

Linking between pages (cont…)

Hyperlinks (cont…)

- Specified using the anchor tag:

  <a href=“destationpath”> name </a>

  - Requires an attribute called “HREF” which specifies the path of the resource to be linked.
- Anchors can be used to go to a particular section in a document.
  - Within the same (or different) document.
  - Example:

    <a href=“face.jpg”> portrait </a>

    <a href=www.google.com> search </a>

    <a href=“mailto:isg@hotmail.com”> Mail me </a>

    <a href=slides/ch5.pdf> download pdf </a>

Linking between pages (cont…)

Relative Vs **Absolute URLs**

- **Absolute URL**
  - The complete path name of the document in the server is specified.
  - Necessary when linking to documents on other servers.
  - Example:
    - <a href="www.iitkgp.ac.in/people/isg/paper5.pdf"> One of my recent papers </a>

Linking between pages (cont...)

Relative Vs **Absolute URLs** (cont...)

- **Relative URL**
  - Provides a link to another document relative to the location of the current document.
  - Commonly used when referring to documents residing on the same web site.
  - Examples:
    - <a href="TENNIS/sania.html"> Sania Mirza </a>
      - means that the document is one folder down from the html document that called for it. This can go on down as many layers as necessary.
    - <a href="../NEWS/cricket.html"> Cricket </a>
      - means that the document is in one folder up from the html document that called for it.

Linking between pages (cont...)

Relative Vs **Absolute URLs** (cont...)

- **Relative URL** (cont...)
  - **Pros:**

    If you upload the whole directory on another server, the links will still work.
  - **Cons:**

    If you copy a part of your web pages on a local disc, the links may not work.

Linking between pages (cont…)

Hyperlinks (Example)

```
<html><head><title> Link to Other Sites </title></head>
  <body >
  My favorite search engines are:
    <ol>
      <li> <a href="https://www.google.com"> Google </a>
      <li> <a href="https://www.facebook.com"> Facebook </a>
      <li> <a href="http://www.bdu.edu.et"> Bahir Dar University</a>
    </ol>
    <hr>
    <address>
      Mr. Ayele Kebede<BR> <BR>
      <a href="mailto:xyz@gmail.com"> Email = xyz@gmail.com </a>
    </address>
  </body></html>
```

Images

- Supported image formats:
  - JPEG, GIF, PNG
- Specified using the standalone tag:

  <IMG src="" height="" width="">
- Attributes of <IMG>:
  - SRC: specifies the URL of the image file
  - HEIGHT: height (in pixels) to be set aside for the image.
  - WIDTH: width (in pixels) to be set aside for the image.

Images (cont…)

- The HEIGHT and WIDTH attributes tells the browser to set aside appropriate space (in pixels) for the image as it downloads the rest of the file.
  - Some browsers stretch or shrink an image to fit into the allotted space.
- Example:

```
<IMG SRC=xyz.gif>
<img src=tiger.jpg height=80 width=150>
```

Images (cont…)

Example 1

```
<html><body>
    An image
    <img src=xyz.gif align="bottom" width="48“ height="48">
    in the text
    <p> An image
    <img src =xyz.gif align="middle" width="48“ height="48">
    in the text
    <p> An image
    <img src =xyz.gif align="top" width="48“ height="48">
    in the text
</body></html>
```

Images (cont...)

Example 2

```
<html>
<body>
    <img src = "xyz.gif" align ="left" width="48" height="48">
    A paragraph with an image. The image will float to the left of this text.
    <p>
    <img src = "xyz.gif" align ="right" width="48" height="48">
    A paragraph with an image. The image will float to the right of this text.
</body>
</html>
```

Images (cont…)

Example 3

```
<html>
<body>
    It is possible to use an image as a link. Click on the image below to go to google.
    <p>
    <a href=“https://www.google.com">
        <img src=x.gif>
    </a>
</body>
</html>
```

HTML Tables

- Tables are made up of cells, arranged into rows.
- Tags used in creating an HTML table:
  - <table></table> marks the start and end of table contents
  - <tr></tr> marks the start and end of each table row
  - <td></td> marks the start and the end of the contents of a data cell
  - <caption></caption> formats text to appear as a table caption

An example:

```
<table>
    <tr>
        <td> Good </td> <td> Bad </td>
    </tr>
    <tr>
        <td> Cute </td> <td> Ugly </td>
    </tr>
</table>
```

Good	Bad
cute	ugly

HTML Tables (cont...)

- <TABLE border=1 bgcolor=blue > ……
  </TABLE>
  - Defines the beginning and the end of a table.
  - Can have several attributes:
    - bgcolor = #rrggbb or color name
    - rules = all | cols | rows | none
    - border = number
    - height = number, percentage

HTML Tables (cont...)

- <TR> …….. </TR>
  - Defines a row of cells within a table.
  - Can have several attributes:
    - bgcolor = #rrggbb or color name
    - align = left | center | right | justify
- <CAPTION> …….. </CAPTION>
  - Provides a caption for the table.
  - Must immediately follow the “table” tag, and precede all other tags.

HTML Tables (cont...)

- <TD> …….. </TD>
  - Defines a table data cell.
  - A table cell can contain any content, like an image, or even another table.
  - Can have several attributes:
    - bgcolor = #rrggbb or color name
    - colspan = number
      - == > specifies the number of columns the currect cell should span (default is 1).
    - rowspan = number
      - == > similar

HTML Tables (cont...)

- <TH> …….. </TH>
  - Defines a table header cell.
  - Browsers generally display the contents of the header cells in bold, and centered.
  - Same attributes as the <TD> tag.

HTML Tables (cont…)

Example 1:

```
<table>
<tr>
<td colspan="2">Hello</td></tr>
<tr><td>good</td><td>day</td></tr>
</table>
```

Hello	
Good	Day

HTML Tables (cont...)

Example 2:

```
<table>
  <tr> <td rowspan=2> Hello </td>
    <td> Good </td> </tr>
  <tr> <td> Day </td> </tr>
</table>
```

Hello	Good
	Day

HTML Tables (cont…)

Example 3:

```
<table border=1>
    <caption> My Table </caption>
    <tr> <th> Name </th> <th> Marks </th> </tr>
    <tr> <td> Ayele</td> <td> 86</td> </tr>
    <tr> <td> Kebede</td> <td> 65</td> </tr>
</table>
```

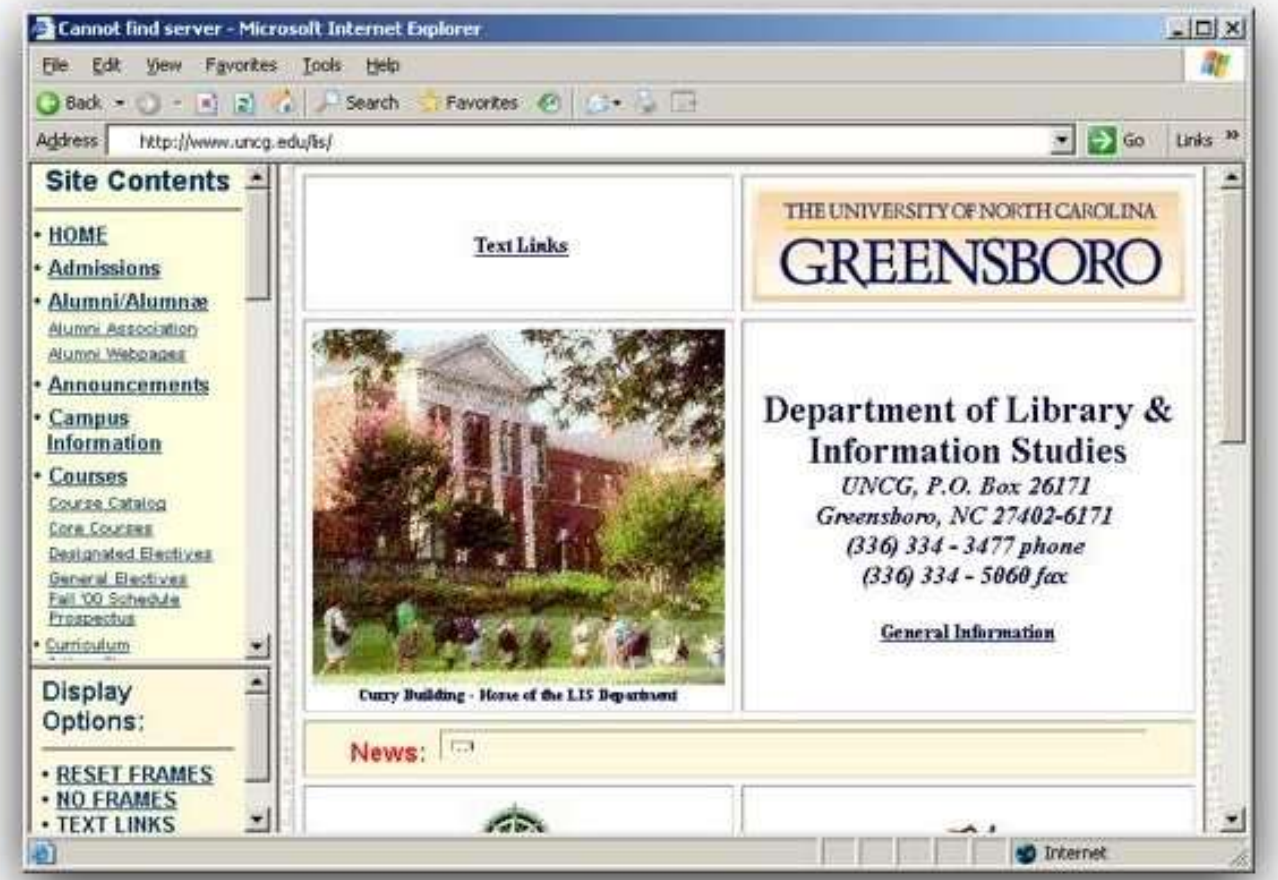
My Table

Name	**marks**
Ayele	86
Kebede	65

HTML Frames

- What are frames?
  - A method for dividing the browser window into smaller sub-windows.
  - Each sub-window displaying a different HTML document.
- How does a page with frame look like?

HTML Frames (cont...)



HTML Frames (cont...)

- Merits:
  - Allow parts of the page to remain stationary while other parts scroll.
  - They can unify resources that reside on separate web servers.
  - There is a <noframe> tag, using which it is possible to add alternative content for browsers that do not support frames.

HTML Frames (cont...)

- Demerits:
  - All browsers do not support frames.
  - Documents nested in a frame is more difficult to bookmark.
  - Load on the server can be significantly increased, if there are a large number of frames in the page.
  - Frames are very difficult to handle for search engines.

HTML Frames (cont...)

<FRAMESET> …….. </FRAMESET>

- Defines a collection of frames or other framesets.
- Attributes:
  - cols = list of lengths (number, %, or *)
  - rows = list of lengths (number, %, or *)
  - Establishes the number and sizes of columns (vertical frames) or rows (horizontal frames) in a frameset.
  - In number of pixels, percentage of total area, or relative values (*) based on available space.

HTML Frames (cont...)

<FRAME>

- Defines a single frame within a frameset.
- Attributes:
  - frameborder = 1 | 0
  - src = url
  - scrolling = yes | no | auto
  - marginwidth = number
  - marginheight = number
  - name = text

HTML Frames (cont...)

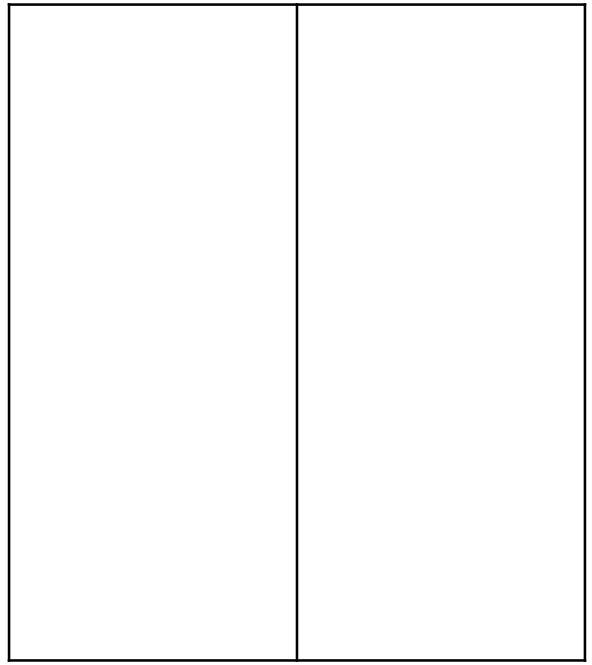
<NOFRAMES> …… </NOFRAMES>

- Specifies the contents to be displayed by browsers that cannot display frames.
- Ignored by browsers that support frames.

HTML Frames (cont...)

Example 1:

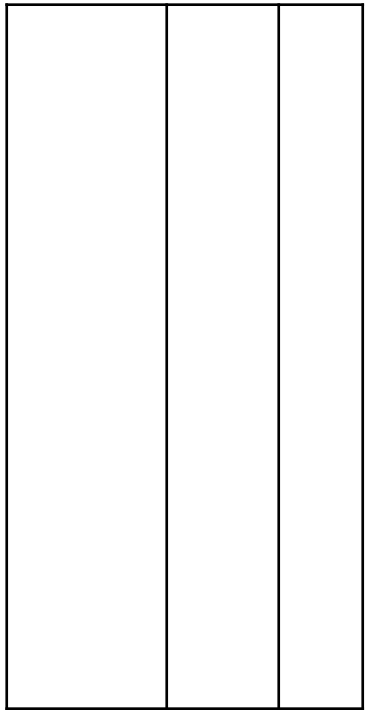
```
<html>
    <head><title> A document with frame </title> </head>
    <frameset rows = "20%,80%">
        <frame src = "left.html">
        <frame src = "right.html">
    </frameset>
</html>
```



HTML Frames (cont...)

Example 2:

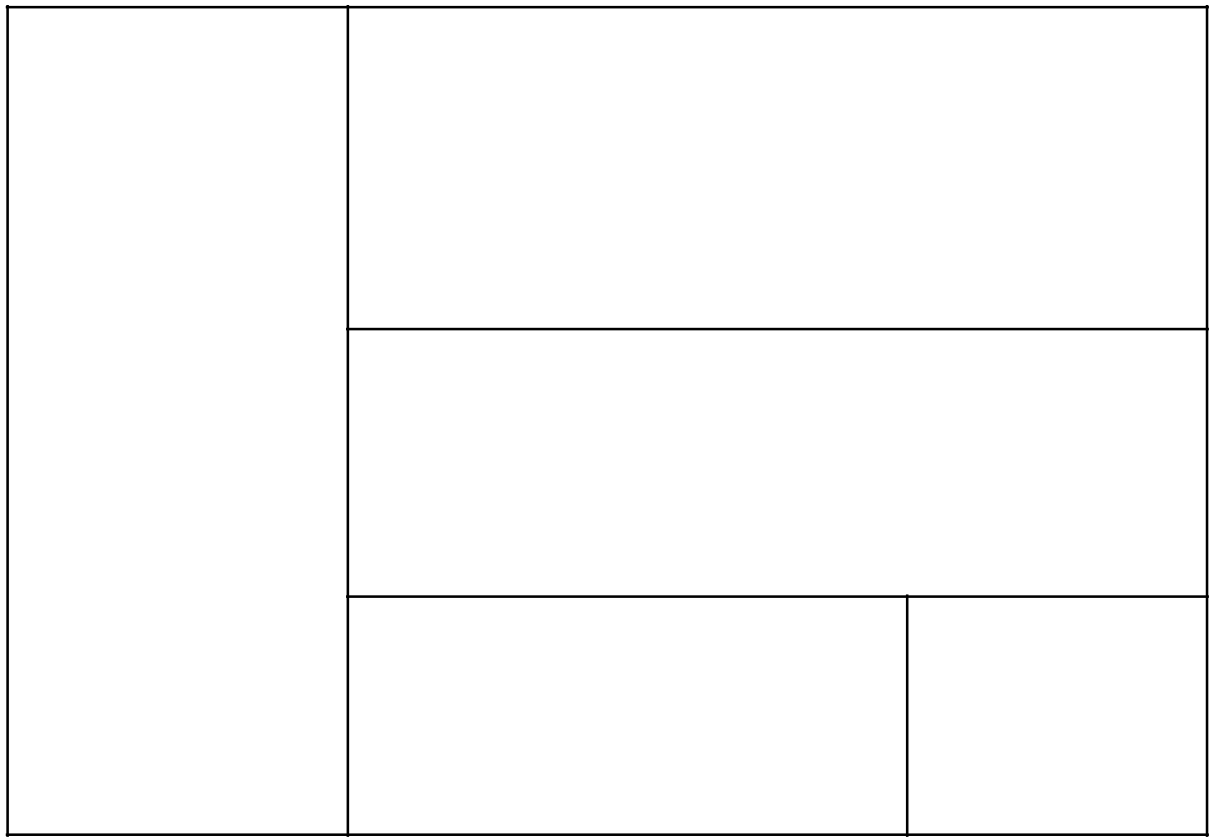
```
<html>
    <head><title> Another one with frames </title> </head>
    <frameset cols = "100,2*,*">
        <frame src = "left.htm">
        <frame src = "right.htm">
    </frameset>
    <noframes>
        Browser does not support frames.
    </noframes>
</html>
```



HTML Frames (cont...)

Example 3:

```
<html>
    <head> <title> Nested frames </title> </head>
        <frameset cols = “25%, *”>
            <frame src = “one.htm”>
        <frameset rows = “100,150,100”>
            <frame src = “two.htm”>
            <frame src = “three.htm”>
        <frameset cols = “*,*”>
            <frame src = “four.htm”>
            <frame src = “five.htm”>
        </frameset> </frameset> </frameset>
</html>
```



HTML Frames (cont...)

Linking to a Framed Document

- When a hyperlink is clicked, by default, the new page is loaded into the same frame.
- We can load the linked page into new frame also.
- Assign a name to the targeted frame.

```
<frame src = "somepage.htm" name = "abc">
```

- Specify this frame in a hyperlink as follows:

```
<a href = "newpage.htm" target="abc"> … </a>
```

- The document newpage.htm will load into the frame names "abc".

HTML Forms

- Provides two-way communication between web servers and browsers.
  - Demand for most of the emerging applications.
  - Providing dynamic contents.
- What is an HTML Form?
  - A form basically contains boxes and buttons.

HTML Forms (cont...)

- Real-life examples
    - Search engines
    - On-line purchase of items
    - Registration
- The form allows a user to fill up the blank entries and send it back to the owner of the page.
    - Called SUBMITTING the form.

HTML Forms (cont...)

FORM Tags and Attributes

- Several tags are used in connection with forms:

```
<form> …… </form>
<input  >
<textarea> …… </textarea>
<select> …… </select>
```

HTML Forms (cont...)

<FORM> …… </FORM>

- This tag is used to bracket an HTML form.
  - Includes attributes which specify where and how to deliver filled-up information to the web server.
- Two main attributes:
  - METHOD
  - ACTION

HTML Forms (cont...)

- METHOD:
  - Indicates how the information in the form will be sent to the web server when the form is submitted.
  - Two possible values:
    - POST: causes a form's contents to be parsed one element at a time.
    - GET: concatenates all field names and values in a single large string.
  - POST is the preferred method because of string size limitations in most systems.

HTML Forms (cont...)

- ACTION:
  - Specifies the URL of a program on the origin server that will be receiving the form's inputs.
  - Traditionally called Common Gateway Interface (CGI).
  - The specified program is executed on the server, when the form is submitted.
    - Output sent back to the browser.

HTML Forms (cont...)

- Typical usage:

```
<FORM METHOD="post“ ACTION="">
……
……
</FORM>
```

HTML Forms (cont...)

<INPUT>

- This tag defines a basic form element.
- Several attributes are possible:
  - TYPE
  - NAME
  - SIZE
  - MAXLENGTH
  - VALUE
  - SRC
  - ALIGN

HTML Forms (cont...)

- TYPE:
  - Defines the kind of element that is to be displayed in the form.
    - “TEXT” – defines a text box, which provides a single line area for entering text.
    - “RADIO” – radio button, used when a choice must be made among several alternatives (clicking on one of the buttons turns off all others in the same group).
    - “CHECKBOX” – similar to the radio buttons, but each box here can be selected independently of the others.

HTML Forms (cont...)

- TYPE (cont...)
  - “PASSWORD” – similar to text box, but characters are not shown as they are typed.
  - “HIDDEN” – used for output only; cannot be modified (mainly used to refer to choices that have already been made earlier).
  - “IMAGE” – used for active maps. When the user clicks on the image, the (x,y) co-ordinates are stored in variables, and are returned for further processing.
  - “SUBMIT” – creates a box labeled Submit; if clicked, the form data are passed on to the designated CGI script.
  - “RESET” – creates a box labeled Reset; if clicked, clears a form’s contents.

HTML Forms (cont...)

- NAME:
  - Specifies a name for the input field.
  - The input-handling program (CGI) in reality receives a number of (name,value) pairs.
- SIZE:
  - Defines the number of characters that can be displayed in a TEXT box without scrolling.
- MAXLENGTH:
  - Defines the maximum number of characters a TEXT box can contain.

HTML Forms (cont...)

- VALUE:
  - Used to submit a default value for a TEXT or HIDDEN field.
  - Can also be used for specifying the label of a button (renaming “Submit”, for example).
- SRC:
  - Provides a pointer to an image file.
  - Used for clickable maps.
- ALIGN:
  - Used for aligning image types.
  - ALIGN = TOP | MIDDLE | BOTTOM

HTML Forms (cont…)

<TEXTAREA> … </TEXTAREA>

- Can be used to accommodate multiple text lines in a box.
- Attributes are:
  - NAME: name of the field.
  - ROWS: number of lines of text that can fit into the box.
  - COLS: width of the text area on the screen.

HTML Forms (cont...)

<SELECT> …. </SELECT>

- Used along with the tag <OPTION>.
- Used to define a selectable list of elements.
  - The list appears as a scrollable menu or a pop-up menu (depends on browser).
- Attributes are:
  - NAME: name of the field.
  - SIZE: specifies the number of option elements that will be displayed at a time on the menu. (If actual number exceeds SIZE, a scrollbar will appear).

HTML Forms (cont...)

<SELECT> …. </SELECT> (cont...)

- MULTIPLE: specifies that multiple selections from the list can be made.

```
<FORM>
  Languages known:
  <SELECT NAME= "lang" SIZE=3 MULTIPLE>
     <OPTION> English
     <OPTION> Amharic
     <OPTION> Afan Oromo
     <OPTION> Tigrigna
  </SELECT>
</FORM>
```

HTML Forms (cont...)

Example 1

```
<HTML>
    <HEAD>
        <TITLE> Using HTML Forms </TITLE>
    </HEAD>
    <BODY TEXT="#FFFFFF"   BGCOLOR="#0000FF" LINK="#FF9900"
      VLINK="#FF9900" ALINK="#FF9900">
    <CENTER><H3> Student Registration Form </H3>
    </CENTER>
        Please fill up the following form about the courses you will register for
          this Semester.
```

HTML Forms (cont...)

```
<FORM METHOD="POST" ACTION="/cgi/feedback">
  <P> Name: <INPUT NAME="name" TYPE="TEXT" SIZE="30" MAXLENGTH="50">
  <P> Roll Number: <INPUT NAME="rollno" TYPE="TEXT" SIZE="7">
  <P> Course Numbers:
      <INPUT NAME="course1" TYPE="TEXT" SIZE="6">
      <INPUT NAME="course2" TYPE="TEXT" SIZE="6">
      <INPUT NAME="course3" TYPE="TEXT" SIZE="6">
  <P> <P> Press SUBMIT when done.
  <P> <INPUT TYPE="SUBMIT">
      <INPUT TYPE="RESET">
  </FORM> </BODY> </HTML>
```

END